VOYAGEX TRAVEL PLATFORM • PROJECT REPORT

Comprehensive Project Implementation Report

Design, Implementation, Security, Containerization & Continuous Deployment

AUTHOR / DEVELOPER	FRAMEWORK & LANGUAGE	DEPLOYMENT CLOUD	GITHUB REPOSITORY
Harshitha R N	Spring Boot 4.x • Java 25	Render (Docker Web Service)	HarshithaR-cyber/voyagex-travelbooking

1. Executive Summary

VoyageX is a modern, enterprise-grade travel and hospitality platform engineered to deliver an integrated travel booking journey encompassing flight discovery, dynamic seat map selection, luxury stays reservation, VIP holiday packages, automated cancellations, and real-time traveler reviews.

The platform is built on an enterprise **Java 25 (LTS)** backend powered by **Spring Boot 4.x** and **Spring Data JPA** with an embedded **H2 Database**. The frontend features a lightweight, high-performance, glassmorphic interface built using modern **HTML5, Vanilla CSS3, and ES6+ JavaScript**. For cloud reliability, the project is packaged into a production-ready multi-stage **Docker container** with automated Continuous Integration & Continuous Deployment (CI/CD) hosted on **Render**.

> **Live Deployment URL:** https://voyagex-travelbooking.onrender.com
> **GitHub Source Code:** https://github.com/HarshithaR-cyber/voyagex-travelbooking

2. Detailed Task Completion Breakdown

The following matrix documents all modules, security workflows, and architectural enhancements engineered throughout the project lifecycle:

Task 1: Mandatory Authentication Gateway & Auth Guard

COMPLETED

Engineered a complete security gating system ensuring visitors must authenticate before entering the VoyageX main portal.

- **Initial Screen Gating:** Root route (`/`) automatically directs unauthenticated visitors to `login.html` before homepage access.
- **Modern Travel-Themed UI:** Designed a responsive, glassmorphic authentication interface consistent with VoyageX branding.
- **Tabbed Authentication Workflow:** Seamless switching between *Sign In*, *Create Account*, and *Forgot Password* forms.
- **Client & Server Validation:** Real-time email syntax regex, password strength checks, and feedback alerts.
- **SessionTokenService:** Cryptographically secure session token generation and server-side validation (`/api/auth/verify-session`).
- **Dynamic Navbar Integration:** Navbar displays authenticated traveler's name, avatar, and active status.
- **Server-Side Logout:** Immediate session revocation on sign-out, clearing client state and returning users to login.

Task 2: Flight Search, Dynamic Pricing & Availability Engine

COMPLETED

Comprehensive flight booking search supporting dynamic origin/destination queries, cabin class tiers, and fare calculation.

- **Multi-Criteria Search:** Origin, destination airport lookup, departure & return dates, passenger counter.
- **Cabin Class Multipliers:** Economy, Premium Economy, Business, and First Class fare structures.
- **Dynamic Pricing Service:** Live price adjustment algorithms calculating taxes, fees, and real-time flight demand.
- **Price Freeze Feature:** Locks quoted flight prices for 48 hours for a small holding fee.

Task 3: Interactive Aircraft Cabin Seat Selection

COMPLETED

Interactive 2D aircraft seating layout enabling passengers to visually choose their preferred seats.

- **Visual Seat Grid:** Real-time visualization of window, middle, and aisle seats.
- **Live State Indicators:** Clear distinction between available, selected, occupied, and extra-legroom seats.
- **Booking Integration:** Selected seat numbers are attached to the passenger's confirmed reservation PNR.

Task 4: Live Flight Status & Tracking Tracker

COMPLETED

Real-time status inquiry for scheduled flights across global departure and arrival gates.

- **Flight Number Query:** Real-time lookup for commercial flight numbers (e.g., VX-101, VX-204).
- **Operational Metadata:** Live reporting on status (On-Time, Boarding, In-Air, Delayed), terminal gates, and baggage carousels.

Task 5: Hotel Recommendations & Interactive Stays

COMPLETED

Hotel discovery and room booking system with personalized filters and amenities.

- **Property Discovery:** Search top accommodations across Paris, Tokyo, Bali, New York, and Dubai.
- **Amenity Badges:** Visual amenity tags for High-Speed Wi-Fi, Swimming Pools, Fitness Centers, and Fine Dining.
- **Room Tier Selection:** Interactive room booking modal with real-time room breakdown and instant confirmation codes.

Task 6: VIP Deals & Curated Luxury Travel Packages

COMPLETED

Premium all-inclusive vacation bundles offering flight + hotel packages with exclusive member savings.

- **Category Filter Tabs:** Quick filtering across Tropical, European, Urban, and Mountain getaways.
- **VIP Perks Highlight:** Dedicated badge highlights for complimentary airport chauffeur, free breakfast, and flexible dates.
- **One-Click Package Booking:** Direct checkout with promotional bundle discounts applied.

Task 7: Booking Cancellation & Transparent Refund Engine

COMPLETED

Fair and automated cancellation system calculating instant refund values based on departure timeline.

- **Automated Refund Calculator:** Transparent tiered deductions based on cancellation lead time.
- **Instant Processing:** Issues official cancellation receipts with transaction tracking IDs.

Task 8: Traveler Reviews & Community Ratings System

COMPLETED

Community-driven traveler review system providing ratings and traveler feedback.

- **Verified Feedback Display:** Dynamic rendering of recent passenger testimonials and star ratings.
- **Review Submission Modal:** Form validation for submitting traveler feedback on hotels and airlines.

Task 9: Cloud Dynamic Port & Docker Containerization

COMPLETED

Engineered multi-stage containerization to achieve standard execution across local and cloud environments.

- **Dynamic Port Configuration:** Configured `server.port=${PORT:8080}` to bind dynamically to cloud-assigned environment ports.
- **Multi-Stage Dockerfile:** Stage 1 builds the JAR with `eclipse-temurin:25-jdk`; Stage 2 packages a slim runtime with non-root security.
- **Optimized .dockerignore:** Excluded unnecessary build files, git histories, and local IDE configs.

Task 10: Git Repository, GitHub Push & Continuous Deployment

COMPLETED

Full version control tracking and cloud automation via GitHub and Render.

- **Git Repository Setup:** Initialized clean Git tree on branch `main` with proper ignore policies.
- **Published to GitHub:** Pushed all source files to `HarshithaR-cyber/voyagex-travelbooking`.
- **Automated Render CI/CD:** Connected Render web service to auto-deploy every commit with zero manual intervention.
- **Comprehensive Documentation:** Delivered production-grade `README.md` and project reports.

3. System Architecture & REST API Specifications

VoyageX is structured using Spring Boot's clean Controller-Service-Repository layered architecture:

METHOD	ENDPOINT	MODULE	DESCRIPTION
POST	`/api/auth/login`	Authentication	Validates credentials & returns session token
POST	`/api/auth/register`	Authentication	Registers traveler account with validation
GET	`/api/auth/verify-session`	Authentication	Validates active session token
POST	`/api/auth/logout`	Authentication	Revokes session token server-side
POST	`/api/auth/forgot-password`	Authentication	Handles password reset requests
POST	`/api/flights/search`	Flights	Searches available flights by criteria
GET	`/api/flights/status/{flightNo}`	Flights	Retrieves live flight tracking information
POST	`/api/bookings/flight`	Bookings	Confirms passenger reservation with seat
GET	`/api/hotels/recommendations`	Hotels	Fetches lodging recommendations by city
POST	`/api/hotels/book`	Hotels	Reserves room with confirmation code
GET	`/api/vip-deals`	VIP Packages	Fetches curated travel packages
POST	`/api/cancellations/quote`	Cancellations	Calculates instant refund quote

4. Verification & Quality Assurance

The platform underwent comprehensive validation to ensure stability, security, and responsive UI rendering:

TEST CASE ID	VERIFICATION SCOPE	ENVIRONMENT	RESULT
VAL-AUTH-01	Unauthenticated redirect to `/login.html`	Integration	PASS
VAL-AUTH-02	Duplicate registration check & password hashing	Unit / Service	PASS
VAL-AUTH-03	Session token generation and lifecycle verification	Unit / Service	PASS
VAL-FLIGHT-01	Flight search criteria filtering & cabin class pricing	API / Controller	PASS
VAL-SEAT-01	Interactive seat selection state toggle and assignment	Frontend E2E	PASS
VAL-HOTEL-01	City-based lodging search & room booking	API / Controller	PASS
VAL-DEAL-01	VIP package filtering by tropical, European, urban tags	API / Controller	PASS
VAL-DOCKER-01	Multi-stage container compilation and minimal JRE image	Docker Engine	PASS
VAL-DEPLOY-01	Dynamic PORT binding & continuous deployment rollout	Render Cloud	PASS

5. Conclusion

The VoyageX travel booking platform successfully meets all architectural, functional, and security objectives. By combining the enterprise capabilities of **Java 25 and Spring Boot** with an intuitive, dynamic frontend and **cloud-native Docker deployment**, the platform delivers a reliable and modern user experience. The integrated GitHub repository and Render CI/CD pipeline ensure that future feature enhancements can be pushed and deployed continuously with zero downtime.